

Differential Expression Analysis with DESeq2

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Outline

From Counts to Questions

The Negative Binomial Model

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Review

Where We Are

- ▶ Yesterday we learned how genomics data is structured: count matrices, `SummarizedExperiment` objects, quality assessment.
- ▶ Today we answer the scientific question that motivated the entire experiment:
Which genes change their expression between treated and untreated cells?
- ▶ This is called **differential expression (DE) analysis**.
- ▶ We will use the airway dataset: human airway smooth muscle cells treated with dexamethasone (a glucocorticoid) vs. untreated controls. Eight samples across four cell lines.

The Statistical Problem

- ▶ For each of $\sim 20,000$ genes, we want to test:

H_0 : mean expression is the same in treated and untreated cells

- ▶ That is 20,000 separate hypothesis tests.
- ▶ Two challenges, both familiar from earlier in the course:
 1. **Multiple testing:** with 20,000 tests at $\alpha = 0.05$, we expect 1,000 false positives by chance alone even if no gene is truly differentially expressed. We need adjusted p-values.
 2. **Small sample sizes:** typical RNA-seq studies have $n = 3-6$ samples per group. Variance estimation from 3 replicates alone is very unreliable.
- ▶ DESeq2 addresses both: it borrows information across genes to stabilize variance estimates, and it uses the Benjamini-Hochberg procedure to control the false discovery rate.

Why Not a t-Test?

- ▶ A two-sample t-test compares means assuming:
 - ▶ Continuous, approximately normal data.
 - ▶ Constant variance across groups.
- ▶ RNA-seq counts violate both:
 - ▶ Counts are non-negative integers, not continuous.
 - ▶ Variance grows with the mean (overdispersion).
- ▶ The appropriate model for RNA-seq counts is the **negative binomial (NB) distribution**:

$$\text{Var}(Y) = \mu + \alpha\mu^2$$

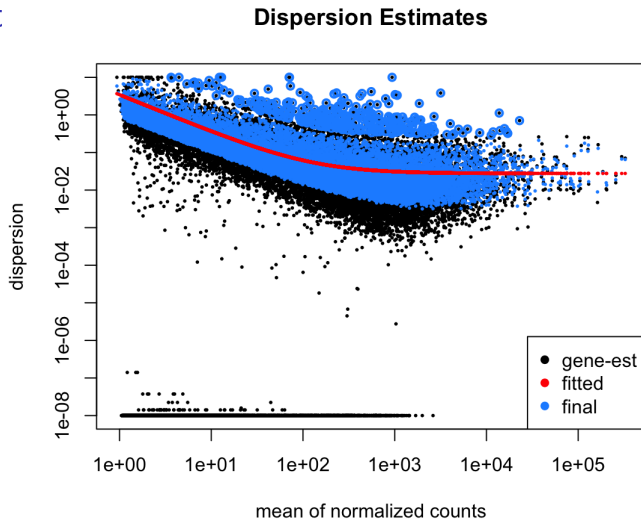
where μ is the mean and $\alpha > 0$ is the **dispersion parameter**.

- ▶ When $\alpha = 0$, the NB reduces to Poisson. RNA-seq data almost always has $\alpha > 0$ (more variable than Poisson).
- ▶ DESeq2 estimates a dispersion parameter for every gene, then tests for differential expression using the NB model.

The Dispersion Challenge

- ▶ With only 3–6 replicates per gene, estimating the dispersion from the data for that gene alone is very noisy.
- ▶ Genes with similar mean expression levels tend to have similar dispersions. DESeq2 exploits this by **shrinking** each gene's dispersion estimate toward a trend fitted across all genes.
- ▶ This is called **empirical Bayes shrinkage** or borrowing strength across genes.
- ▶ The result: more reliable variance estimates for every gene, even those with only 3 replicates.
- ▶ The dispersion plot (produced by `plotDispEsts()`) shows the per-gene estimates (black dots), the fitted trend (red line), and the final shrunken estimates (blue dots).

Dispersion Plot



Normalization for Sequencing Depth

- ▶ Raw counts are not directly comparable across samples because samples differ in **sequencing depth** (total reads).
- ▶ DESeq2 estimates a **size factor** for each sample: a single number that captures how deeply that sample was sequenced relative to the others.
- ▶ The method: for each gene, compute its geometric mean across all samples. Each sample's size factor is the median ratio of its counts to those geometric means.
- ▶ Normalized counts = raw counts divided by the size factor.
- ▶ You do not compute this manually. DESeq2 does it internally when you call `DESeq()`, and stores the size factors in the `DESeqDataSet` object.
- ▶ You can inspect them with `sizeFactors(deseq_dds)`.

The DESeq2 Workflow in Five Steps

1. **Build the DESeqDataSet.** Provide counts, sample metadata, and the design formula.
2. **Filter low-count genes.** Remove genes with very few total reads across all samples. These carry no information and slow computation.
3. **Set the reference level.** Tell DESeq2 which group is the control so that fold changes are in the right direction.
4. **Run DESeq().** One function call estimates size factors, dispersions, and fits the NB model for every gene.
5. **Extract and interpret results.** `results()` returns a table with one row per gene: log2 fold change, p-value, and adjusted p-value.

In R : steps 4 and 5 are almost trivially simple. The work is in understanding what the output means.

The Results Table

`results(deseq_dds)` returns a table with these columns:

Column	Meaning
baseMean	Average normalized count across all samples
log2FoldChange	$\log_2(\text{treated}/\text{untreated})$
lfcSE	Standard error of the log2 fold change
stat	Wald test statistic
pvalue	Raw p-value
padj	Adjusted p-value (Benjamini-Hochberg)

- ▶ `padj` is the primary measure of significance. Use it, not `pvalue`.
- ▶ A positive `log2FoldChange` means the gene is **upregulated** in treated cells.
- ▶ A negative `log2FoldChange` means **downregulated**.

Log2 Fold Change: The Effect Size

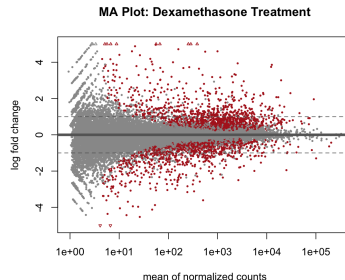
- ▶ The **log2 fold change (LFC)** tells you the direction and magnitude of the expression change.
- ▶ Three reference points:
 - ▶ $\text{LFC} = 0$: no change in expression.
 - ▶ $\text{LFC} = 1$: treated cells have twice the expression ($2^1 = 2$).
 - ▶ $\text{LFC} = -1$: treated cells have half the expression ($2^{-1} = 0.5$).
- ▶ In practice, LFC values of ± 1 or greater are often considered biologically meaningful, but this is context-dependent.
- ▶ **Caution:** genes with very low mean expression can have extreme LFC estimates even when the evidence is weak. This is why we look at padj , not just LFC.

Adjusted P-values and the FDR

- ▶ Testing 20,000 genes at $\alpha = 0.05$: we expect $0.05 \times 20,000 = 1,000$ false positives by chance.
- ▶ The **false discovery rate (FDR)** is the expected proportion of significant results that are false positives.
- ▶ DESeq2 uses the **Benjamini-Hochberg (BH) procedure** to control the FDR. If we call all genes with $\text{padj} < 0.05$ significant, we expect at most 5% of those calls to be false positives.
- ▶ This is different from the FWER (family-wise error rate) controlled by Bonferroni: BH is less conservative and finds more true positives at the cost of a slightly higher false positive rate.
- ▶ Multiple testing is covered in more depth on Day 13.

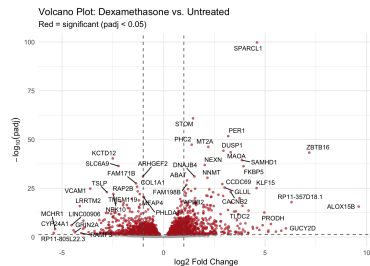
The MA Plot

- ▶ An **MA plot** puts:
 - ▶ **M** ($\log_2$ fold change) on the y-axis.
 - ▶ **A** (mean expression) on the x-axis.
- ▶ Each point is one gene.
- ▶ Significant genes ($p_{adj} < 0.05$) are highlighted.
- ▶ **What to look for:**
 - ▶ Most genes cluster near $LFC = 0$ (no change).
 - ▶ Low-expression genes $\Rightarrow$ wide LFC scatter.
 - ▶ Significant genes should not be concentrated only at very low expression levels (that would suggest a filtering problem).
- ▶ In R : `plotMA(res, ylim = c(-5, 5))`



The Volcano Plot

- ▶ A **volcano plot** combines:
 - ▶ **Effect size** ($\log_2$ fold change) on the x-axis.
 - ▶ **Significance** ($-\log_{10} p_{adj}$) on the y-axis.
- ▶ Genes in the top corners are the most interesting: large effect and high significance.
- ▶ Genes near the bottom have low significance regardless of their fold change.
- ▶ Genes in the middle have small effect sizes regardless of their significance.
- ▶ The volcano plot is exploratory. It is for identifying candidates to follow up, not for making formal claims.



Heatmap of Top DE Genes

- ▶ A **heatmap** of the top differentially expressed genes visualizes the expression pattern across all samples simultaneously.
- ▶ Rows are genes, columns are samples. Color encodes expression level (typically on a log or VST scale).
- ▶ What to look for:
 - ▶ Samples from the same treatment group should cluster together (columns).
 - ▶ Upregulated genes should show high values across treated samples and low values across untreated, and vice versa.
 - ▶ A clean block pattern (two groups of genes, two groups of samples) is the expected result of a strong treatment effect.
- ▶ In R : use the `pheatmap` package or the `ComplexHeatmap` package for full control over annotations.

Adjusting for Covariates in DESeq2

- ▶ Just as in logistic regression, we can include additional variables in the DESeq2 model to adjust for confounders.
- ▶ The **design formula** is where this happens.
- ▶ In the airway experiment, four different cell lines were used. If we do not account for cell line, biological differences between cell lines inflate the residual variance and reduce power.
- ▶ Compare two designs:
 - ▶ `design = ~ dex` — treatment only.
 - ▶ `design = ~ cell + dex` — adjust for cell line, then test treatment.
- ▶ Adding `cell` as a covariate removes its contribution from the residual variance, making the test for `dex` more powerful.
- ▶ The interpretation of the result is the same: the LFC for `dex` now represents the treatment effect *after accounting for cell-line differences*.

Review: The DE Analysis Workflow

1. Build a `DESeqDataSet` with the count matrix, sample metadata, and design formula.
2. Filter low-count genes. Set the reference level.
3. Run `DESeq()` – one call does normalization, dispersion estimation, and model fitting for every gene.
4. Extract results with `results()`. Report `padj`, not `pvalue`. Positive LFC = upregulated; negative LFC = downregulated.
5. Visualize with MA plot, volcano plot, and heatmap of top genes.
6. If the experiment has blocking variables or covariates (cell line, batch, sex), include them in the design formula.

Coming up (Day 13): why the Benjamini-Hochberg procedure works, how to interpret the FDR, and what to do when the number of significant genes is very large or very small.